

SCQ (Single Correct Type) :

1. If $S_n = \sum_{r=1}^n t_r = \frac{1}{6}n(2n^2 + 9n + 13)$, then $\sum_{r=1}^{\infty} \frac{1}{r \cdot \sqrt{t_r}}$ equals

- (A) 1 (B) 2 (C) $\frac{3}{2}$ (D) $\frac{1}{2}$

2. If $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $P = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, $Q = P^T A P$, find $PQ^{2014}P^T$:

- (A) $\begin{pmatrix} 1 & 2^{2014} \\ 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 1 & 4028 \\ 0 & 1 \end{pmatrix}$ (C) $(P^T)^{2013} A^{2014} P^{2013}$ (D) $P^T A^{2014} P$

3. If $\Delta(x) = \begin{vmatrix} x-1 & 5x & 7 \\ x^2-1 & x-1 & 8 \\ 2x & 3x & 0 \end{vmatrix} = ax^3 + bx^2 + cx + d$ then the value of $a+2b+3c+5d$ is

- (A) 154 (B) 150 (C) 156 (D) 149

MCQ (One or more than one correct) :

4. Let a, x, b be in A.P; a, y, b be in G.P and a, z, b be in H.P. If $x = y + 2$ and $a = 5z$, then

- (A) $y^2 = xz$ (B) $x > y > z$ (C) $a = 9, b = 1$ (D) $a = 1/4, b = 9/4$

5. Let $A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} a \\ b \\ 1 \end{bmatrix}$. If AB is a scalar multiple of B , then

- (A) $4a + 7b + 5 = 0$ (B) $a + b + 2 = 0$
 (C) $b - a = 4$ (D) $a + 3b = 0$

6. A is a square matrix of order 3. A, A^{-1}, A^T all has the same value of determinant. Also $(\text{adj}A) = A^T$, then

- (A) $\det(A^{-1}) = 1$
 (B) $(ABA^T)^2 = A^2 B^2 (A^T)^2$, where B is a 3×3 matrix
 (C) $\det \{(ABA^T)^{1000}\} = (\det B) \Rightarrow \det(B) = 0, 1 \text{ or } -1$
 (D) $(ABA^T)^{-1} = AB^{-1} A^T$, if B is an invertible 3×3 matrix

7. Let A, B, C be 3 non-singular square matrix of same order, such that $A^T = A, B^T = B$ also $CAB = CBA$, then

- (A) $AB^{-1} = B^{-1}A$ (B) AB is symmetric (C) $ABA^{-1} = B$ (D) $BAB^{-1} = A$

8. z_1 and z_2 are two complex numbers satisfying $i|z_1|^2 z_2 - |z_2|^2 z_1 = z_1 - iz_2$. Then which of the following is/are correct?
- (A) $\operatorname{Re}\left(\frac{z_1}{z_2}\right) = 0$ (B) $\operatorname{Im}\left(\frac{z_1}{z_2}\right) = 0$
- (C) $z_1 \bar{z}_2 + \bar{z}_1 z_2 = 0$ (D) $|z_1| |z_2| = 1$ or $|z_1| = |z_2|$
9. $\sum_{r=1}^{1006} |z^{2r+1} - z^{2r-1}| = \sum_{r=1}^{1006} |z^{2r} - z^{2r-2}| = 2012$, where z is a complex number, then (given $i^2 = -1$)
- (A) $|z| = 1$ (B) $|z^2 - 1| = 2$ (C) $z = \pm i$ (D) $z = \frac{1 \pm i}{\sqrt{2}}$

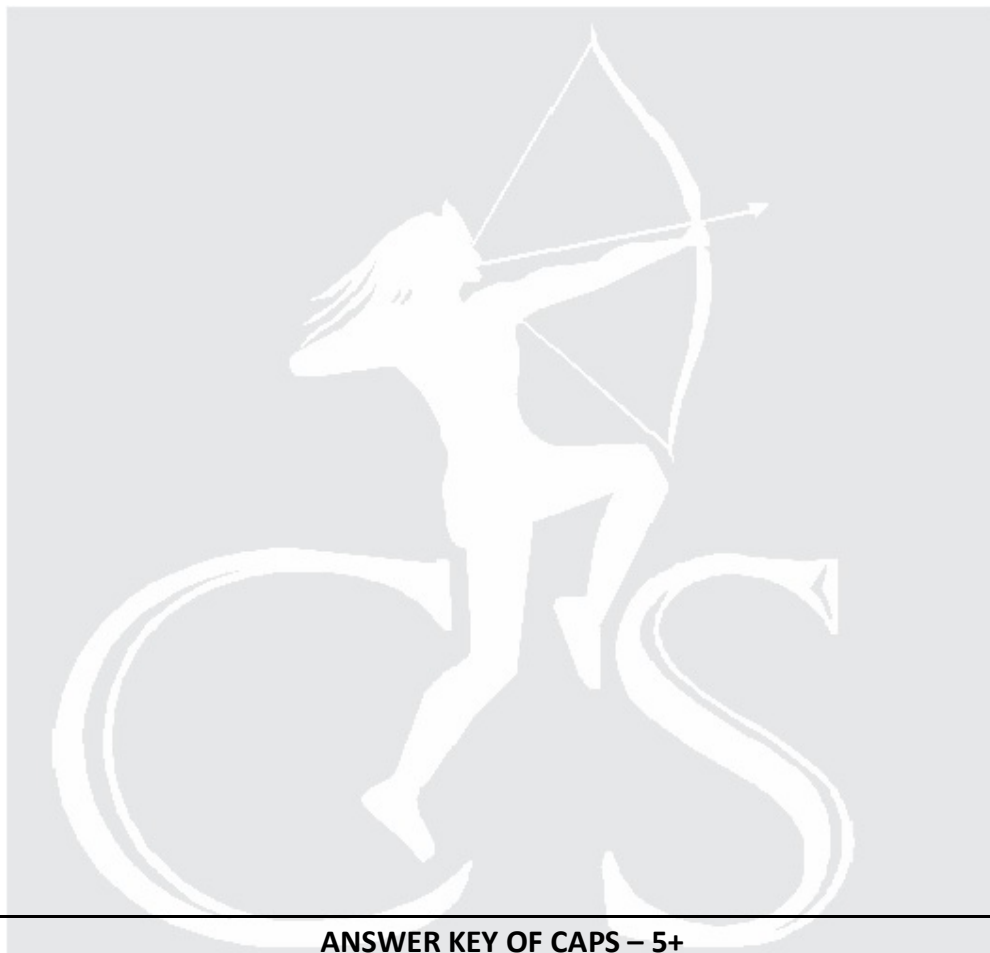
Numerical Based Questions :

10. If $\sum_{k=1}^{\infty} \frac{k^2}{3^k} = \frac{p}{q}$; where p and q are relatively prime positive integers. Find the value of $(p + q)$.
11. A and B are two square matrices. Such that $A^2 B = BA$ and if $(AB)^{10} = A^k \cdot B^{10}$. Find the value of $k - 1020$.
12. Let $A = \begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$, where $x, y, z \in \mathbb{N}$, If $\det \{\operatorname{adj}(\operatorname{adj}(\operatorname{adj}(A)))\} = 4^8 5^{16}$ then number of such matrix A is _____
13. If $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$, $B = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$
- also $(A + A^2 B^2 + A^3 + A^4 B^4 + \dots + 100 \text{ terms}) = (k)A$, where $k \in \mathbb{R}$, find k _____
14. A function 'f' is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex number z , where α and γ are complex numbers if $f(1)$ and $f(i)$ are both real and the smallest possible values of $|\alpha| + |\gamma|$ is p then determine p^2 .
15. If ω is imaginary root of $z^{28} = 1$ and such that $|\omega + 1|$ is maximum. If $x = \frac{1}{2} \left| \omega - \frac{1}{\omega} \right|$ then $8x^4 + 4x^3 - 8x^2 - 3x + 4$ is equal to :

Subjective Type Problems :

16. The points represented by the complex numbers a, b, c lie on a circle with centre O and radius r . The tangent at c cuts the chord joining the points a, b at z . Show that $z = \frac{a^{-1} + b^{-1} - 2c^{-1}}{a^{-1}b^{-1} - c^{-2}}$
17. Find the radius and the centre of the circle which follows $\operatorname{Arg}\left(\frac{z-1-2i}{z+3+i}\right) = \frac{\pi}{3}$
18. Show that for the given complex numbers z_1 and z_2 and for a real c the equation $(z_1 + \lambda z_2)\bar{z} + (\bar{z}_1 + \lambda \bar{z}_2)z + c = 0$ represents a family of concurrent lines and also find the fixed point of the family.

19. If z_1 and z_2 are the two complex numbers satisfying $|z - 3 - 4i| = 8$ and $\text{Arg}\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then find the range of the values of $|z_1 - z_2|$.
20. Let a, b, c be distinct complex numbers such that $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$, ($a, b, c \neq 1$). Find the value of k .



ANSWER KEY OF CAPS – 5+

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|---|----------|----------|---|-----------|----------|
| 1. (A) | 2. (B) | 3. (C) | 4. (ABC) | 5. (ABCD) | 6. (ACD) |
| 7. (ABCD) | 8. (ACD) | 9. (ABC) | 10. 5 | 11. 3 | 12. 36 |
| 13. 100 | 14. 2 | 15. 3 | 16. $z = \frac{c^2(a+b) - 2abc}{c^2 - ab} = \frac{a^{-1} + b^{-1} - 2c^{-1}}{a^{-1}b^{-1} - c^{-2}}$ H.P. | | |
| 17. $\left[\left(-1 - \frac{2}{\sqrt{3}} \right) + i \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right) \right]$ | | | 18. $z = \frac{Cz_2}{z_1\bar{z}_2 - z_2\bar{z}_1}$ | | |
| 19. $ (z_1 - z_2) \in [3\sqrt{2}, 13\sqrt{2}]$ | | | 20. $-\omega$ or $-\omega^2$ | | |